Supplementary Information

Endpoint and Exposure Sensitivity in Fault Assessment of a Fly-Connectome Spiking Model

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Review package assessment-v8, 22 October 2026

This document is the current entry point for all supporting measurements. It consolidates the evidence retained across protocols v2–v4 and v8; their status statements describe earlier revisions and are not current experimental claims. This revision adds 48 fault episodes to 24 preserved healthy episodes, completing the 72-record controller-by-fault comparison in Sections 12–13. The matched artifact contains raw records, generating-source snapshots where recorded, current analysis scripts, figure sources and a SHA256/size manifest.

1 Campaign and material index

Tables S1–S22 and Figures S1–S9 distinguish historical observations from new simulations. Main-paper Table I reports the controller-by-fault matrix; Tables S12–S14 retain the earlier healthy-only intervention evidence. The +30-degree healthy direction is at floor in both prospective cohorts. Neural survival endpoints are bounded at 1000 ms; the geometric rule uses the separate horizons in Table S1.

Table S1: Campaigns and statistical units. Counts refer to neural episodes, except the deterministic rule.

Campaign	Seeds	Bearings	Runs	Duration (ms)
Passive calibration	0–2	$-60, -30, 0$	9	400
First prospective campaign	6–11	$-30, +30$	48	1000
Paired healthy controls	12–17	$-30, +30$	48	1000
Controller-by-fault extension	12–17	$-30, +30$	48 new + 24 reused	1000
Sensory refractory sensitivity	12–14	Constant drive	12	600
Main historical sweep	Fault-seeded	Constant drive	19×12	600
Privileged geometric rule	Deterministic	$-30, +30$	6	800/1000/1200

The 288 scored fault/control records comprise 216 main-sweep fault trials plus 72 targeted/random-inhibitory controls; healthy references are separate. The target-permuted historical phase is a specificity control, not a functioning fault-tolerance baseline. Exact current tables derive from saved summaries; no missing historical manifest or generation version is reconstructed as if it had been recorded.

Reading map. Section 2 defines model and estimands; Section 3 supplies every main dose; Section 4 gives cohorts, controls and legacy partitions; Section 5 separates stopping-time matching from late crossings; Sections 6–7 give the prospective and paired-control results; Section 8 gives model-variant sensitivity and compute; Sections 9–10 give archived diagnostic figures; Section 11 gives reproduction and provenance boundaries; Sections 12–13 add the paired factorial experiment. Tables and figures are numbered continuously and have explicit sample/dispersion definitions.

2 Model, endpoints and statistical definitions

The signed FlyWire-v783 substrate contains 138,639 neurons, 15,091,983 connection rows and approximately 54.5 million contacts. Membrane and synaptic-drive constants are 20 and 5 ms; threshold/reset are -45/-52 mV, delay 1.8 ms, and contact gain 0.275 mV. Both voltage equations pause during the 2.2-ms refractory period. Project wrappers retain this interval in sensory neurons, unlike the upstream stimulation routine. Fresh integration uses 0.1 ms; historical timestep metadata are unavailable. Independent 150-Hz inputs drive 104 LC4 and 210 LPLC2 cells; each external event adds 68.75 mV to voltage. Eight DNp01/DNa02/DNp04/DNb01 cells provide the readout. No background drive is used.

Open-loop endpoints. For eight command cells and $T = 0.6$ s, $r = N_{\mathrm{cmd}}/(8T)$. Latency is the first spike in this group, not a completed decision. Total S counts all network spikes. Transmission requires $r \geq 51.4$ Hz, rounding $(94.583333 + 8.229167)/2 = 51.40625$ Hz from separate ten-trial sensory/random-drive calibrations. Joint passing additionally requires $r \leq c\,94.6875$ Hz and $S \leq c\,39694$ with $c = 2$; sensitivity uses $c = 1.5, 3, 4$. This is a nested operational rule, not a validated energy/safety budget.

Paired estimands. With cumulative spikes S_F, S_H and recorded durations T_F, T_H, define

$$R_{\mathrm{whole}} = \frac{S_F(T_F)}{S_H(T_H)}, \quad R_{\mathrm{common}} = \frac{S_F(T_*)}{S_H(T_*)}, \quad R(t) = \frac{S_F(t)}{S_H(t)}, \quad T_* = \min(T_F, T_H),\ t \leq T_*.$$

Matching unequal stopping times and choosing an earlier endpoint are different. The common duration can itself depend on fault-induced collision. Ratios are reported per scenario before taking medians. Undefined zero-reference ratios are not silently assigned finite values. Legacy timestamps mark frame starts but cumulative counts include the completed 20-ms frame; 400 ms uses frame 19.

Frozen controller. Nine passive episodes at -60/-30/0 degrees and seeds 0–2 fix threshold 18.125 Hz, bias $b = -20$ Hz, scale $\ell = 70$ Hz and sign +1. The fitted window is 250–300 ms. Steering is $\omega_k = -400\,\mathrm{deg/s}\ \mathrm{clip}((\bar{d}_k - b)/\ell, -1, 1)$. d_k sums four left rates minus four right rates; a_k is the eight-cell mean. Bars average the latest five frames (all available initially). Triggering requires the current smoothed $\bar{a}_k$ and preceding unsmoothed a_{k-1} both to reach 18.125 Hz. The first qualifying frame schedules one burst 250 ms after its end. This is not two consecutive smoothed crossings. Burst speed/duration are 0.35 m/s and 450 ms, with fixed heading; otherwise speed is 0.05 m/s and heading is limited to ±100 degrees. Cue onset is 100 ms. Geometry uses actual separation, body-frame bearing and within-frame collision checks at 0.05 m.

Uncertainty. Historical median intervals use 20,000 resamples, seed 0. Reference-aware ratio-of-median intervals independently resample healthy/fault trials. Dispersion bars instead show 10th–90th trial percentiles. Legacy seed diagnostics enumerate $3^3 = 27$ ordered draws, yielding only 4–7 distinct medians. New control ranges resample six complete seed clusters, keeping both bearings together. These conditional percentile ranges are exploratory, with no confirmatory coverage, significance or equivalence claim. More bootstrap draws do not add independent observations. Historical open-loop streams were not reset per trial.

3 Complete archived main sweep

Table S2: All main conditions, 12 trials each. Rate, latency and spikes are medians; brackets give conditional 95% bootstrap intervals for rate.

Condition	Rate [range] (Hz)	Latency (ms)	Spikes	T
F1_wvar_05	94.38 [93.75,95.94]	4.20	39,678.5	12/12
F1_wvar_10	95.31 [94.17,95.52]	4.20	39,723.0	12/12
F1_wvar_20	94.17 [92.71,96.67]	4.20	39,988.0	12/12
F1_wvar_50	94.17 [92.19,96.46]	4.35	41,638.0	12/12
F2_drop_05	91.88 [91.15,93.75]	4.40	39,077.0	12/12
F2_drop_10	93.12 [90.62,96.04]	4.30	38,448.0	12/12
F2_drop_25	84.48 [82.29,87.81]	4.65	36,598.0	12/12
F2_drop_50	72.40 [71.25,74.69]	5.40	31,589.5	12/12
F3_loss_01	94.58 [93.85,95.00]	4.30	39,335.0	12/12
F3_loss_02	94.27 [93.44,95.83]	4.15	39,936.0	12/12
F3_loss_05	94.06 [92.71,95.73]	4.10	39,373.5	12/12
F3_loss_10	95.10 [93.65,96.56]	4.15	39,753.5	12/12
F4_hub_1pct	119.06 [118.33,119.90]	4.20	49,272.5	12/12
F4_hub_2pct	114.79 [114.69,115.21]	4.20	38,861.0	12/12
F4_hub_5pct	93.96 [93.75,94.27]	4.25	34,115.5	12/12
F5_in_both	0.00 [0.00,0.00]	—	21,191.0	0/12
F5_in_LC4	75.52 [75.10,76.15]	6.10	34,543.5	12/12
F5_in_LPLC2	64.27 [63.75,64.90]	5.00	27,686.0	12/12
healthy	94.69 [94.06,95.42]	4.05	39,694.0	12/12

F1 multiplies each weight by $\exp(\sigma Z - \sigma^2/2)$, $Z \sim N(0,1)$; $\sigma = 0.5$ has coefficient of variation 53.3%. F2 independently deletes connection rows, so expected lost absolute contact mass is the deletion fraction. F3/F4 silence outgoing connections while affected cells can still receive input and spike. F5 silences sensory outputs. Historical F4 selections lack contemporaneous cell manifests and may include protected readouts. These rows are descriptive, not corrected-selector results.

The ratio of F2_50 and healthy median rates is 0.764576. Resampling both groups gives a 95% bootstrap interval [0.751101, 0.789648]. The first-spike-latency ratio is 1.333333 [1.253012, 1.400000]. These ratios of medians differ from the median of within-seed ratios in Table S15. A separate control at 75 Hz with 50% deletion passes 10/12; the 150-Hz sweep does not establish drive invariance or equivalence.

4 Cohorts, realized controls and legacy partitions

Table S3: Cohort scoring at $c = 2$. T and J denote transmission and joint passes.

Included records	n	T	J	T−J
All fault/control records	288	276	262	14
Exclude inhibitory hubs	276	264	262	2
Exclude F4/HUB/EIm	204	192	190	2
Main F1/F2/F3/F5	180	168	168	0

Table S4: Historical controls, 12 trials each. Medians of recorded damage and responses; cell/edge counts are not contact-mass equality.

Condition	Cells	Rows modified (%)	Rate (Hz)	Spikes
EIm_1pct	—	1.13	94.48	40,087.5
EIm_2pct	—	2.25	95.10	39,882.5
HUB_exc	—	3.48	88.65	32,265.0
HUB_inh	—	5.57	168.23	847,476.5
INHrand_cells	765	0.75	95.21	39,827.0
INHrand_mass	8,889	8.63	102.40	50,323.0

EIm denotes excitatory/inhibitory-composition-matched controls. The two remaining additional rejections after targeted-hub exclusion are INHrand_mass trials 3 and 9 (zero-based), with 327,246 and 271,058 spikes. Their median cannot summarize this tail. Complete per-trial damage is preserved in the raw JSONL files; dashes denote absent cell-count metadata, not zero cells. Post-hoc selector reconstructions are supplied at `audit/verified-hubs` with their original manifest. They cannot establish missing historical hub identities; expected-mass matching does not imply realized contact equality.

Table S5: Healthy legacy partitions. Exact reused logs differ from shared seed clusters.

Partition	Scenarios	Collision-free
Calibration reuse	$-60/-30$ degrees, seeds 0–2	6/6
Other negative bearings	$-60/-30$ degrees, seeds 3–5	6/6
Positive bearings	$+30/+60$ degrees, seeds 0–5	12/12
Frontal	0 degrees, seeds 0–2	0/3

All legacy fault records use calibration seed clusters. The remaining 18/18 healthy records lack exact scenario reuse but do not form an independent evaluation.

5 Stopping times, early windows and late crossings

Table S6: Median paired legacy spike ratios, using main-paper Eq. (3). Ratios are computed within scenario before aggregation.

Condition	Whole	Common duration	First 400 ms
F2_25	0.8861	0.8861	0.8837
F2_50	0.3817	0.7294	0.7226
F4_1pct	1.7029	1.7029	1.3618
HUB_inh	71.6931	75.2306	118.1571

Table S7: Full-hub cumulative horizon sensitivity. Every fault/healthy pair lasts 1000 ms.

Horizon (ms)	Median paired ratio	Above twofold
200	1.302	0/12
300	1.332	0/12
400	1.362	0/12
500	1.402	0/12
600	1.415	0/12
700	1.435	0/12
800	1.472	0/12
900	1.682	6/12
1000	1.703	6/12

Table S8: First twofold crossings. All six are negative-bearing calibration scenarios.

Bearing (degrees)	Seed 0 (ms)	Seed 1 (ms)	Seed 2 (ms)
-60	900	880	880
-30	840	840	860
30	None	None	None
60	None	None	None

For F2_50, common-duration matching changes 0.381701 to 0.729443. For full hubs, whole and common ratios are identical within every pair. Selecting 400 ms excludes late activity, without invalidating the one-second warning. Inhibitory hubs exceed twofold in all 12 pairs at every selected 200–500-ms window. Late-crossing traces are in main-paper Figure 3; Figure S3 shows individual estimands rather than only pooled medians.

6 First prospective campaign: directional outcomes

Table S9: First campaign, seeds 6–11. Six clusters per condition; bearings share seeds.

Condition	−30 degrees	+30 degrees	Total	Burst at 1 s
Healthy	5/6	0/6	5/12	3
25% deletion	4/6	0/6	4/12	2
50% deletion	0/6	0/6	0/12	0
Readout disabled	0/6	0/6	0/12	0

Table S10: First-campaign collision-free counts over both bearings within each seed (maximum two).

Condition	6	7	8	9	10	11
Healthy	0	1	1	1	1	1
25% deletion	0	1	0	1	1	1
50% deletion	0	0	0	0	0	0
Readout disabled	0	0	0	0	0	0

Table S11: Median paired activity in first-campaign faults. All 12 pairs are retained in each cell.

Condition	200 ms	300 ms	400 ms	500 ms
25% deletion	0.9103	0.9357	0.9832	1.0073
50% deletion	0.7484	0.8451	0.8898	1.0123

Neither deletion condition exceeds twofold matched activity in these windows despite eight and twelve collisions. This is not a fitted task-failure detector. The activity ratio is distinct from the open-loop joint criterion. Every +30-degree healthy record collides, creating a floor for fault comparisons in that direction. Five healthy survivors are retained at -30 degrees; 25% deletion loses one and 50% loses all five on those paired seeds.

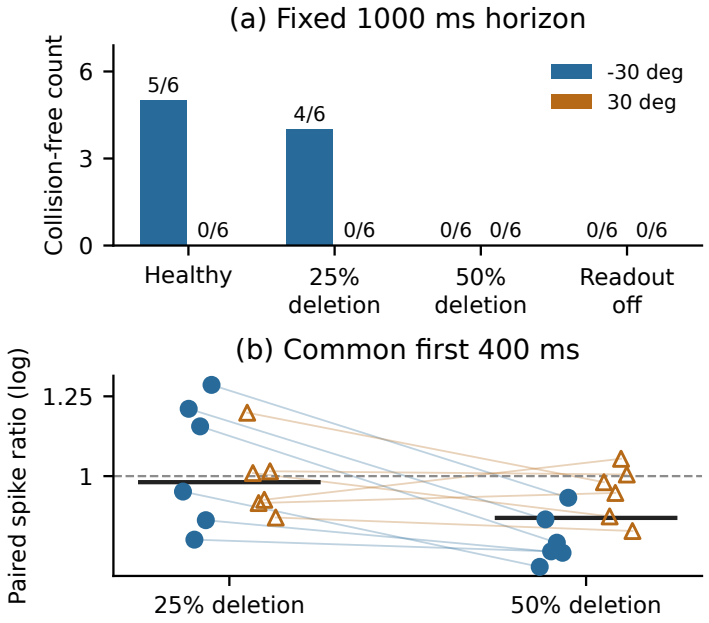


Figure S1: First-campaign outcomes and every first-400-ms pair. Lines join the same seed and bearing across doses; directions share six seed clusters. Dashed ratio one is a reference; black segments are medians.

7 Paired controller interventions: new seeds

Table S12: Healthy-network controls, seeds 12–17. All retain the same neural trigger rule, not identical trigger times.

Law	−30 degrees	+30 degrees	Total	Bursts triggered
Original law	4/6	0/6	4/12	4/12
Pre-cue gate	6/6	0/6	6/12	6/12
Offset only	0/6	0/6	0/12	11/12
No steering	0/6	0/6	0/12	12/12

Table S13: Every paired seed outcome. Each cell lists (-30,+30) binary survival at 1 s.

Seed	Original law	Pre-cue gate	Offset only	No steering
12	1,0	1,0	0,0	0,0
13	1,0	1,0	0,0	0,0
14	0,0	1,0	0,0	0,0
15	1,0	1,0	0,0	0,0
16	0,0	1,0	0,0	0,0
17	1,0	1,0	0,0	0,0

Table S14: Mean paired difference versus original law, averaging both bearings within seed. Conditional 95% percentile bootstrap ranges, 20,000 six-cluster draws; exploratory only.

Law	Difference (pp)	Range (pp)	Improved / worse / same
Pre-cue gate	+16.7	[0.0, 33.3]	2 / 0 / 4
Offset only	-33.3	[-50.0, -16.7]	0 / 4 / 2
No steering	-33.3	[-50.0, -16.7]	0 / 4 / 2

The gate comparison has two improving, zero worsening and four unchanged clusters. Its range includes zero. All positive-bearing tests collide under every law. Offset-only and no-steering laws trigger bursts in 11/12 and 12/12 cases yet never survive. These complete feedback-law comparisons do not isolate trigger timing from subsequent sensory trajectories. They do not estimate deletion tolerance under altered control laws by themselves; Sections 12–13 supply the direct fault comparison. Original/gated survivors still bursting at 1 s number four/three. Every group's survival count is unchanged between 800 and 1000 ms; no 1200-ms neural outcome is imputed.

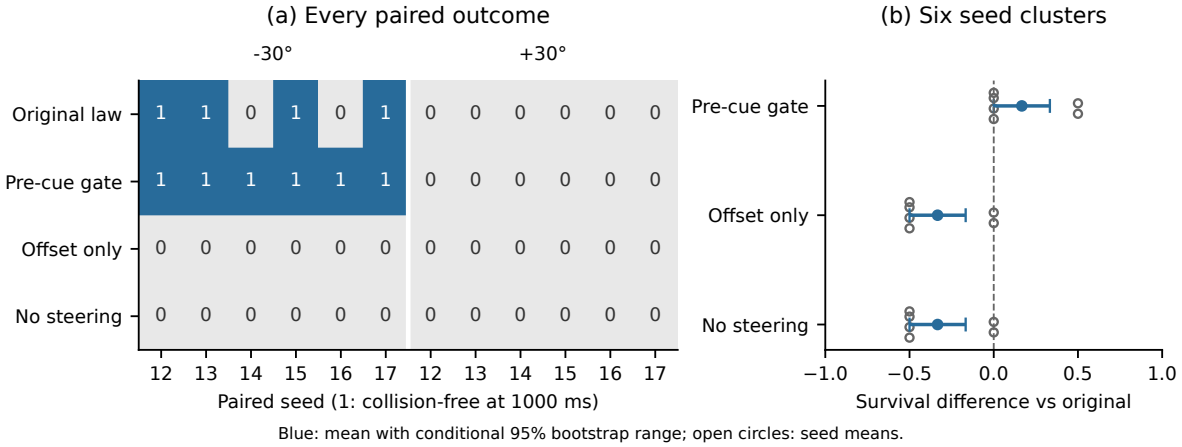


Figure S2: Every paired outcome and six-cluster differences. Blue intervals are conditional exploratory percentile ranges; open circles are seed means. Bearings are fixed scenarios sharing each seed.

8 Sensory refractory sensitivity and measured compute

Table S15: Three-seed model-variant comparison, 12 full-connectome 600-ms runs. Retention is computed within seed before its median; it is not a ratio of group medians.

Sensory refractory	Healthy (Hz)	Fault (Hz)	Paired median	Paired range
2.2 ms	94.79	72.50	0.7632	[0.7451, 0.8285]
0.0 ms	103.12	81.67	0.7919	[0.7619, 0.8525]

Seeds 12–14 share deletion masks (10000+s) and reset Poisson seeds (20000+s) within each pair. All other neurons retain 2.2-ms refractoriness. Healthy median rates are 94.79 and 103.12 Hz under sensory refractory times 2.2 and 0 ms. This quantifies a conditional implementation difference; three paired seeds cannot establish variant invariance or replace the historical sweep.

Table S16: Measured compute on Intel Xeon W-2223, 3.60 GHz, Windows. Sums of worker episode wall times are not CPU time.

Campaign	Episodes	Elapsed (s)	Episode sum (s)
Paired controls, 3 workers	48	2134.5	6256.5
Refractory, 1 worker	12	1222.4	1219.7

Refractory construction took 6.2 s before its elapsed interval. Control timings include state restoration and frame processing; workers overlap.

Fresh neural execution uses CPython 3.13.15, Brian2 2.10.1, NumPy 2.5.4 and the NumPy backend, with 0.1-ms integration and 20-ms feedback. Observed per-process working memory is about 1.1 GB, not a profiled peak bound. Total historical campaign runtime is unavailable. A complete Cython/compiler environment has not been validated. The original 48-test campaign has generation hashes but not every complete version-specific source snapshot. New control and refractory batches save exact generating Python sources with per-file hashes.

Assessment boundary. One connectome, selected drives, asymmetric calibration coverage, assumed receptive fields and directional floor effects limit transfer. Fixed-horizon survival is not eventual escape. The privileged geometric rule survives both bearings at 800/1000 ms but collides by 1200 ms; it establishes bounded feasibility, not a matched neural resource-cost baseline. No new hardware mapping, corrected-hub causal contrast, predictive activity budget, biological escape model or population reliability is established.

9 Supplementary visual diagnostics

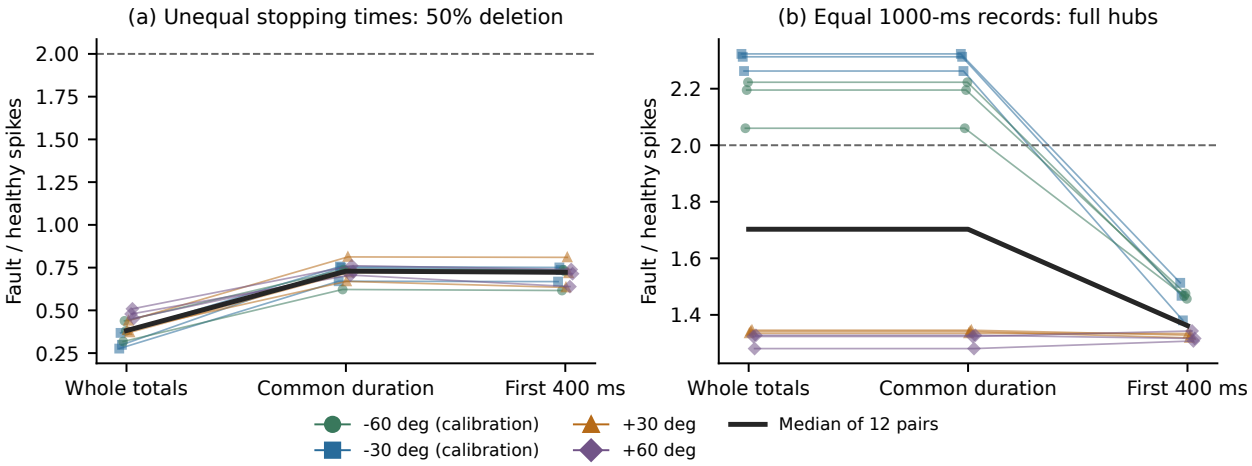


Figure S3: Individual paired whole/common/400-ms estimands. Shape and color identify bearing; negative-bearing pairs reuse calibration scenarios, while positive bearings still share calibration seeds. The black line is the median of all 12 pairs.

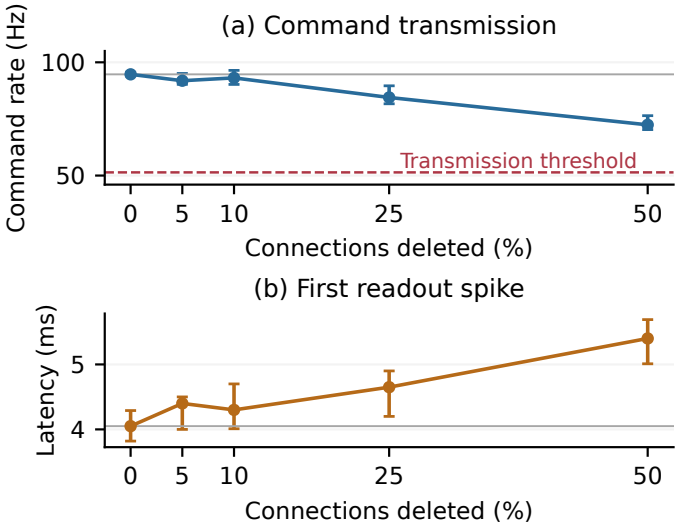


Figure S4: Archived deletion doses, 12 trials each. Points are medians; bars show observed 10th–90th percentiles, not confidence intervals. Healthy references and the 51.4-Hz transmission threshold are separate lines.

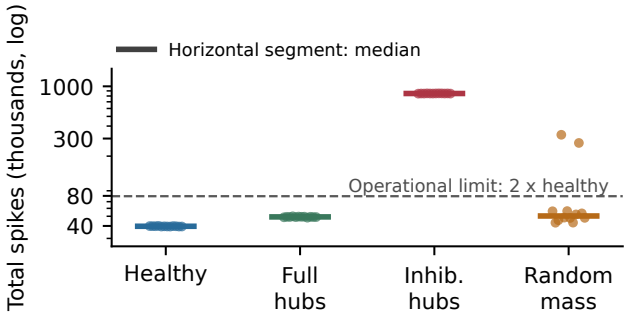


Figure S5: Historical network spikes in thousands, logarithmic scale. Each dot is a trial; horizontal segments are medians. The dashed twofold limit is operational, not an energy or safety budget. Both random-mass tail trials are retained.

10 Legacy diagnostics and reproduction

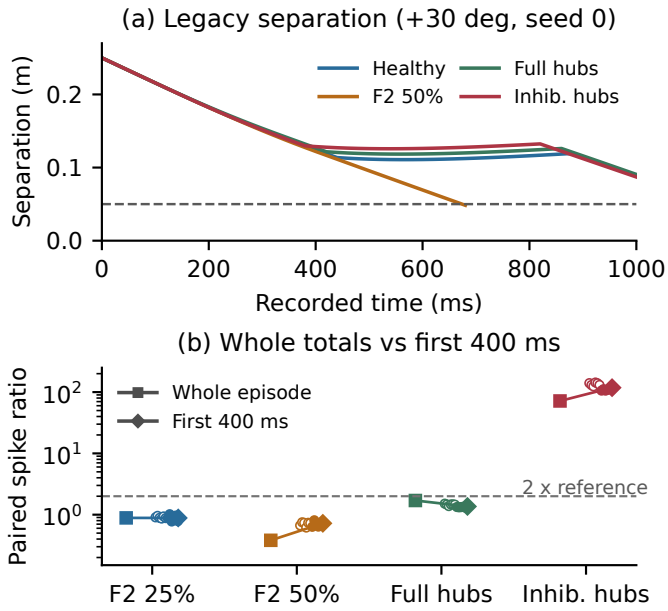


Figure S6: Legacy illustration, not a test of corrected geometry. The separation trace uses +30 degrees, seed 0. Squares/diamonds are whole/400-ms medians; dots show 400-ms pairs, open for calibration scenarios. Whole records can have unequal durations. Full-hub equal-duration late crossings remain valid in main Figure 3.

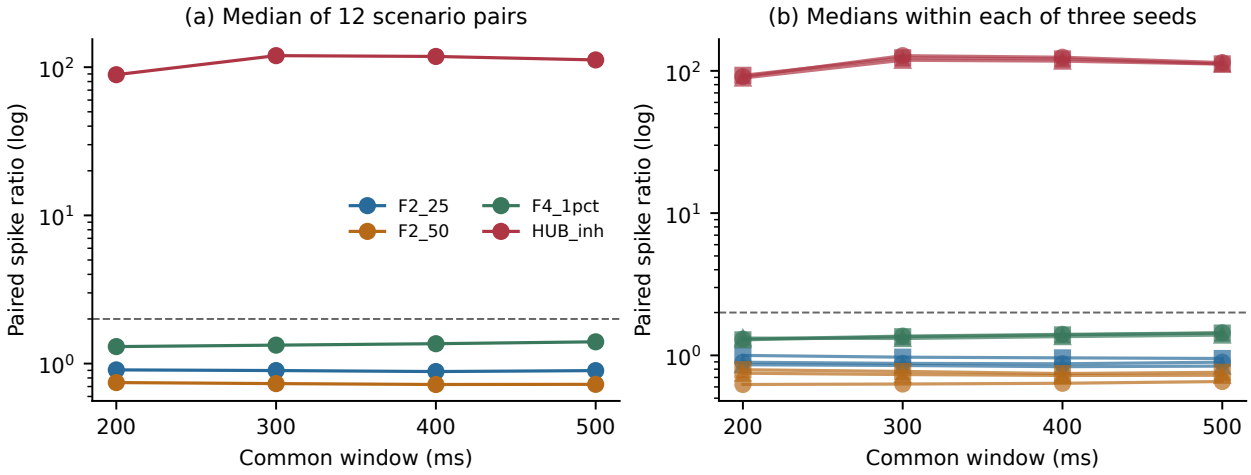


Figure S7: Post-hoc early-window diagnostics, all 12 scenario pairs and three shared seed medians. The 500-ms endpoint does not cover the full-hub late crossings. Refer to main Figure 3 for the complete one-second time course.

11 Reproduction and exact artifact access

The review package contains `main.pdf`, this `supplement.pdf`, a material index and the matching source/data ZIP. The current revision is prepared locally; it has not been submitted or published. The repository link in the paper alone does not expose this revision. For review, the current files must be supplied through a permitted submission mechanism or an explicitly authorized accessible release. A hash check is evidence of file identity, not public availability or acceptance.

Read-only reconstruction. From the repository root, install the declared analysis dependencies in a CPython environment, then use each campaign's `analyze()` function to compare against its saved JSON. Entry points are

- `tools/analyze_controls_v4.py`
- `tools/analyze_horizons_v4.py`
- `tools/analyze_refractory_v4.py`
- `tools/summarize_prospective.py`

The scripts refuse accidental overwrite of their original output directories. Raw reconstructions require no new neural simulation. Exact build commands, versions and expected external-input hashes are in `REPRODUCE.md`.

Source and data identity. The artifact manifest lists every included file by SHA256 and size, and binds both PDFs to their sources. External `Connectivity_783.parquet` and `Completeness_783.csv` tables remain separate dependencies from the released Shiu model repository. Expected hashes and Windows newline normalization are documented; no table values were changed. The calibration file remains byte-identical across both prospective campaigns. No final-release source hash is substituted for a missing older generation hash. The bundled upstream source snapshots retain their license notice.

Reporting procedure supported by this case study. Report the cohort denominator and condition composition; identify the functional and activity criteria separately; state the paired stopping rule; inspect full-horizon activity; and report task outcomes by direction and independent seed cluster. These reporting choices clarify evidence without establishing a safety predictor.

Prior work and inherited components. The manuscript bibliography supplies the full verified citations. The neural substrate follows Shiu et al., Nature 2024, DOI 10.1038/s41586-024-07763-9. Close comparisons include An's sparsification study, DOI 10.64898/2026.09.19.752860, and Jo's endpoint-dependent damage study, arXiv:2610.19153. NeuroBench, DOI 10.1038/s41467-025-56739-4, already motivates separate functional/system measures. Our contribution is the quantified case study of cohort, exposure, horizon and feedback-law dependence; neither adding a nested constraint nor defining a ratio is claimed as an algorithmic invention.

Historical documentation. Protocols v2–v4 and earlier repair reports are retained for chronology. Their old pending-work statements are superseded by this current supplement and its campaign index. Historical raw records remain immutable; only current summaries, presentation and delivery files are revised.

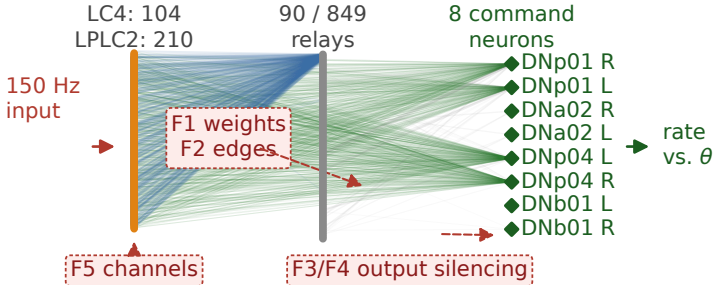


Figure S8: Mapped pathway retained from the previous main Figure 1. For display, 90 of 849 relays are selected by summed sensory-input and command-output contacts. Labels follow mapped identities; line strength encodes contacts and dashed markers locate perturbations. Generator and input hashes accompany the artifact. This display does not establish a dynamical mechanism; the new main Figure 1 is a conceptual overview.

12 Controller-by-fault comparison: frozen calibration

This incremental campaign adds 48 actual full-connectome fault episodes and reuses 24 byte-identical healthy episodes from the paired controls. The six seed clusters (12–17), both bearings, calibration, receptive-field map, neural sources and package versions are shared across all six conditions. Healthy outcomes were already known when this extension was specified; the fault outcomes were not used to retune the controller. Each seed uses Brian2 seed $20000 + s$ and deletion seed $10000 + s$. F2 deletes weighted connection-table rows, including all contacts represented by each row. The same row mask is used across controllers and bearings. The 25/50-percent masks are nested draws, not independent fault realizations.

Let Y_H, Y_F denote 1000-ms collision-free survival under the same law, bearing and seed. Loss is $L = I(Y_H = 1, Y_F = 0)$; gain is $G = I(Y_H = 0, Y_F = 1)$; net loss is $D = Y_H - Y_F = L - G$. Loss counts are out of all 12 scenarios; the healthy survivors at risk are also reported. Both bearings and all six conditions remain together within each seed cluster. Thus 72 records do not represent 72 independent observations.

Table S17: Complete factorial outcomes. L/G: paired losses/gains; H: healthy survivors at risk. Healthy rows are reused, not new runs.

Law	Deletion	-30°	$+30^\circ$	Total	L	G	H
Original	Healthy	4/6	0/6	4/12	—	—	—
Original	25%	4/6	0/6	4/12	1	1	4
Original	50%	0/6	0/6	0/12	4	0	4
Pre-cue gate	Healthy	6/6	0/6	6/12	—	—	—
Pre-cue gate	25%	3/6	0/6	3/12	3	0	6
Pre-cue gate	50%	0/6	0/6	0/12	6	0	6

Table S18: Every scenario outcome: each cell lists binary survival (-30,+30 degrees). H denotes healthy; 25/50 denote deletion percentages.

Seed	Original H	Original 25	Original 50	Pre-cue gate H	Pre-cue gate 25	Pre-cue gate 50
12	1,0	0,0	0,0	1,0	0,0	0,0
13	1,0	1,0	0,0	1,0	0,0	0,0
14	0,0	1,0	0,0	1,0	1,0	0,0
15	1,0	1,0	0,0	1,0	1,0	0,0
16	0,0	0,0	0,0	1,0	0,0	0,0
17	1,0	1,0	0,0	1,0	1,0	0,0

Table S19: Every paired loss/gain count within seed (two bearings). A gain is retained rather than discarded from the net loss.

Seed	Original 25	Original 50	Gate 25	Gate 50
12	1,0	1,0	1,0	1,0
13	0,0	1,0	1,0	1,0
14	0,1	0,0	0,0	1,0
15	0,0	1,0	0,0	1,0
16	0,0	0,0	1,0	1,0
17	0,0	1,0	0,0	1,0

13 Paired interaction, activity and threshold sensitivity

The interaction is $D_{\text{gate}} - D_{\text{original}}$, after averaging both bearings within seed. Positive values mean greater net fault loss under gating. Conditional 95-percent percentile ranges use 20,000 complete six-cluster bootstrap draws with RNG seed 0. These exploratory ranges are conditional on the frozen calibration and fixed bearings. They do not establish population coverage, equivalence or a controller mechanism. The +30-degree healthy floor limits observable loss in that direction.

After observing the conditional bootstrap ranges, we added a post-hoc sign-flip sensitivity, applying all $2^6 = 64$ sign patterns to the six seed-level interaction contrasts. Under an independent-cluster, symmetric zero-centered interaction null, the two-sided tail fraction is 0.25 at 25-percent deletion (three nonzero clusters) and 0.50 at 50-percent deletion (two nonzero clusters). This diagnostic is not a randomized controller-assignment test or confirmatory population inference. The positive percentile range at 25 percent must be interpreted with this sparse support; it is not a claim of established controller-dependent neural resilience. At 50 percent, both faulty controllers have zero survivors, so the net-loss contrast reflects healthy baseline success.

Table S20: Controller interaction, in percentage points (pp). Six seed values are the actual independent cluster contrasts, not six extra experiments.

Deletion	Six seed contrasts (pp)	Mean (pp)	Range (pp)
25%	0, 50, 50, 0, 50, 0	+25.0	[8.3, 41.7]
50%	0, 0, 50, 0, 50, 0	+16.7	[0.0, 33.3]

Table S21: Median within-scenario fault/healthy network-spike ratios under the same law. Each cell retains all 12 pairs. Common duration is pair-specific and outcome-dependent.

Law / deletion	200 ms	300 ms	400 ms	500 ms	Common
Original 25%	0.8889	0.8976	0.9532	0.9623	0.9712
Original 50%	0.7293	0.7525	0.9074	0.9969	1.0322
Pre-cue gate 25%	0.8863	0.9414	0.9457	0.9470	0.9701
Pre-cue gate 50%	0.7525	0.7750	0.8785	0.9275	1.0796

Table S22: Activity-only flags R(t)>c versus collision/survival through 1000 ms. FC/UC: flagged/unflagged collisions; FS/US: flagged/unflagged survivors. Counts for c=2,3,4 are identical and combined. All four 200–500-ms windows appear in thresholds.csv.

Law / deletion	t (ms)	c	FC	UC	FS	US
Original 25%	400	1.5	0	8	0	4
Original 25%	400	2/3/4	0	8	0	4
Original 25%	500	1.5	0	8	1	3
Original 25%	500	2/3/4	0	8	0	4
Original 50%	400	1.5	0	12	0	0
Original 50%	400	2/3/4	0	12	0	0
Original 50%	500	1.5	0	12	0	0
Original 50%	500	2/3/4	0	12	0	0
Pre-cue gate 25%	400	1.5	0	9	0	3
Pre-cue gate 25%	400	2/3/4	0	9	0	3
Pre-cue gate 25%	500	1.5	0	9	1	2
Pre-cue gate 25%	500	2/3/4	0	9	0	3
Pre-cue gate 50%	400	1.5	0	12	0	0
Pre-cue gate 50%	400	2/3/4	0	12	0	0
Pre-cue gate 50%	500	1.5	0	12	0	0
Pre-cue gate 50%	500	2/3/4	0	12	0	0

These flags use network spikes only; they are distinct from the archived open-loop joint rule, which also requires

command transmission. The thresholds are operational definitions, with no validated hardware resource, power or safety interpretation. No threshold was selected to fit task outcomes. There is no imputation of neural activity after collision. Activity pairs use the same controller, bearing and seed at completed common frames.

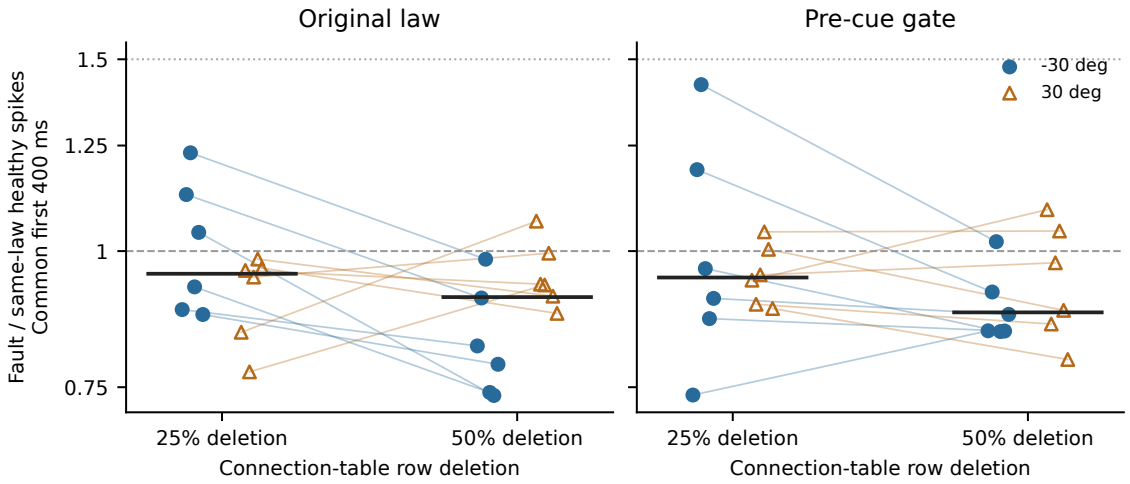


Figure S9: Every paired seed/bearing activity ratio at 400 ms. Lines join the same seed and bearing across nested deletion doses within a control law. Color/shape identify direction; thick black bars are medians. Both bearings share six seed clusters.

The successful resumed invocation took 1727.1 s, including worker construction and skipping completed records. This excludes the interrupted invocation. The sum of measured episode times across all 48 new fault episodes is 6162.4 s; overlapping workers mean this is not CPU time or total campaign elapsed time.

All three resumed workers returned zero. Each completed episode was written before interruption recovery; the runner verified source/configuration identity and did not overwrite it. The source and calibration hashes are checked independently by `tools/cross_records_v8.py`. Machine-readable per-scenario loss, gain, four fixed windows and common duration are in `audit/controller-fault-v8/pairs.csv`; complete summaries and manifests are in `audit/controller-fault-v8/cross.json`. Protocol details and commands are in `paper/PROTOCOL_v8.md`. Corrected historical hub experiments and additional test bearings remain outside this focused revision.